

IPv4 and ARP

4/15/26 2:31 PM

Summary:

- **Layer 2 (MAC/ARP):** Local, easily spoofed → monitor switch logs and **ARP** for **MITM** or reconnaissance.
- **Layer 3 (IPv4):** Routable but spoofable → focus on **TTL anomalies**, source validation, and abnormal traffic.
- **Detection workflow:**
 1. Capture traffic (**Wireshark** / **tcpdump** / **Suricata**).
 2. Filter anomalies (**ARP** floods, low/varying **TTL**, unusual protocols).
 3. Correlate with host artifacts (processes, logs).
 4. Block suspicious sources and investigate root cause.

Common Tools & Commands:

- **Wireshark:** Packet inspection (arp, ip.ttl, eth.src filters).
- **tcpdump:** Command-line capture (tcpdump -e -nn -r file.pcap).
- **arp -a:** View **ARP** cache.
- **Suricata:** Rule-based detection of **ARP** storms or **IPv4** anomalies.

These fundamentals help you quickly spot **Layer 2/3** attacks in real incidents (e.g., **ARP** poisoning during data exfil or **TTL** changes indicating **MITM**).

1. MAC Addressing (Layer 2)

What is a MAC address? 48-bit hardware identifier for network interfaces (e.g., **00:1A:2B:3C:4D:5E**). First 3 bytes = manufacturer (**OUI**), last 3 bytes = device-specific.

Security relevance: MAC is used only for local network communication. It is **not routable** and **easily spoofed**. Attackers change MAC to evade tracking, bypass filters, or impersonate devices.

Spoofing MAC Addresses (common attack):

- Attacker changes their MAC to look trusted or hide identity.
- Linux example:

```
Bash
sudo ip link set eth0 down
sudo ip link set eth0 address 00:11:22:33:44:55
sudo ip link set eth0 up
```

Detection in IR:

- Sudden **MAC changes** for the same IP (check switch logs or ARP tables).
- Multiple devices claiming same MAC → spoofing or conflict.
- **Wireshark filter:** eth.src == xx:xx:xx:xx:xx:xx.

Key takeaway: Never trust **MAC** as a security control. Treat it as untrusted in investigations.

2. Address Resolution Protocol (ARP)

What ARP does: Maps **IP address** (Layer 3) to **MAC address** (Layer 2) on the local network.

Normal ARP flow:

1. Host checks **ARP cache** (arp -a).
2. If missing, broadcasts **ARP Request** (destination MAC = all **FF:FF:FF:FF:FF:FF**).
3. Target replies with unicast **ARP Reply** containing its MAC.
4. Sender updates cache and sends data.

ARP Attacks:

- **ARP Poisoning (MITM):** Attacker sends fake ARP Replies claiming wrong MAC for an IP. Traffic is redirected through attacker.
- **ARP Storm/Flood:** Attacker floods network with ARP requests → DoS (overloads switches/hosts). One MAC pretending to own many IPs.

Detection:

- **ARP storm:** Sudden spike in ARP requests (**Wireshark filter** arp or Suricata rules).
- **Poisoning:** Same IP mapped to multiple MACs or inconsistent replies.

Demo: ARP Storm Detection:

- Use **tcpdump** or **Wireshark** on PCAP.
- Command example:

```
Bash
tcpdump -e -nn -r arp-storm.pcap | cut -d' ' -f2,10 | cut -d'.' -f1-4 | sort |
uniq -c | sort -n
```

- Look for one **MAC** associated with many IPs (anomalous).

Key takeaway: **ARP** has no authentication. Monitor for floods or poisoning in any suspicious network incident.

3. IPv4 and Malicious Traffic

IPv4 Basics: 32-bit logical address (e.g., **192.168.1.10**). Header includes **Source IP**, **Destination IP**, **TTL**, Protocol, etc. Unlike MAC, **IPv4** is routable across networks.

Time To Live (TTL):

- Starts at OS default (**Windows = 128**, **Linux = 64**, routers often **255**).
- Decrements by 1 per router hop.
- Reaches 0 → packet dropped (prevents loops).
- **Security use:** Baseline normal TTL. Anomalies help detect **spoofing**, **tunneling**, or **MITM** (extra hop).

Detecting Anomalies Using TTL (Demo):

- In **Wireshark:** Add **TTL** column or filter ip.ttl < 50.
- Normal: Consistent high TTL for same source.
- Suspicious: Sudden drop (e.g., from 47 to 46) or unexpectedly low TTL → possible extra device (**MITM**) or spoofing.
- Example: Client packet **TTL=128** (Windows), server reply **TTL=47** → suggests ~17 hops. Sudden change to 46 = investigate as potential **MITM**.

Other Malicious IPv4 Patterns:

- **IP spoofing** (fake source IP).
- **Fragmentation attacks** (overlapping fragments to evade IDS).
- Abnormal packet sizes or unusual protocols from one source.

Detection & IR Response:

- Use **ACLs/firewalls** to validate source IPs.
- Monitor **TTL anomalies** and spoofed traffic.
- Correlate with host logs (e.g., unexpected processes making outbound connections).
- In **Wireshark:** Filter by source IP and watch **TTL** changes.

Security Best Practices:

- Understand your network’s normal routing behavior.
- Block unexpected traffic patterns.
- Move toward better segmentation and **IPv6** where possible (stronger security features).

Key takeaway: **IPv4** has weak built-in security. Use **TTL**, header inspection, and traffic patterns to spot anomalies like spoofing, tunneling, or hidden C2 during incident response.